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Flow Experience Analysis in Gameful Educational Environments: A Systematic Literature Review

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***Abstract*—One of the most important challenges in gameful educational environments (*e.g.*, gamified educational systems, educational games, and educational VR applications) is to identify students' experience (*e.g.*, engagement, motivation, and flow experience) during the system usage. Especially regarding flow experience, in recent years, different studies have analyzed students' flow experience while using gameful educational environments. However, it is difficult to identify the state-of-the-art flow experience analysis in gameful educational environments. Facing this challenge, we present a systematic literature review investigating how the flow experience analysis is conducted in gameful educational environments. The main results indicate that most of the studies are using psychometrically validated scales to measure the students' flow experience. Our results especially contribute to the fields of educational technologies and gameful design in education, presenting a state-of-the-art flow experience analysis in gameful educational environments.**

***Index Terms*—Flow Theory, gameful environments, game-based learning, user experience, systematic literature review**

I. Introduction

In the realm of gameful educational environments (*e.g.*, gamified educational systems, educational games, and educational virtual reality applications), a pivotal concern revolves around understanding and analyzing students' experience (*e.g.*, engagement, motivation, and flow experience) [1]. Especially, the flow experience (*i.e.*, holistic sensation that people feel when they act with total involvement [2]) that is characterized by deep engagement and optimal immersion [2], is considered a crucial factor in enhancing learning outcomes and motivation within educational settings [3]–[5]. For this reason, over the last few years, several studies have analyzed the students' flow experience in gameful educational environments [6].

Amidst the rapid evolution of gameful educational approaches, identifying the experience of flow in such environments (*i.e.*, identifying whether a learner achieved the flow experience while using a gameful educational environment) has become an increasing challenge [7]–[9]. Especially with the development of new technologies (*e.g.*, augmented reality glasses and smartphones), it has become increasingly complex to understand how the flow experience has been analyzed in gameful educational systems [10]–[12]. At the same time, current literature fails to provide secondary studies that allow the community to understand the state-of-the-art in the analysis of users' flow experience in gameful educational environments.

To address this challenge and provide comprehensive insights into flow experience analysis, this study employs a systematic literature review examining a wide array of works (*i.e.*, 131 studies elected for the full reading), encompassing empirical studies aiming to present a comprehensive overview of the current landscape of flow experience analysis within gameful educational environments. Thus, we answered the following research question: how the flow experience is analyzed in gameful educational environments?

The main findings of our analysis reveal the majority of studies have primarily relied on the use of scales as measurement tools for assessing students' flow experience, and a significant emphasis has been placed on employing psychometrically validated instruments, underscoring the importance of robust measurement approaches in capturing the intricacies of flow in educational settings. We contribute to different fields, *e.g.*, educational technologies, gameful design in education, and Positive Psychology, by offering a comprehensive overview of flow experience analysis in gameful educational environments.

II. Method

The systematic literature review was conducted based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol [13]. In this section, we present the steps followed in the study (following the PRISMA protocol). All subjective definitions of the process (*e.g.*, definition of the search string and data extraction) were made by two researchers with about 10 years of experience in gameful studies, Flow Theory, and in conducting secondary studies.

The main objective of this study was to comprehensive overview of flow experience analysis in gameful educational environments. To achieve this objective, we answered the following research question: how the flow experience is analyzed in gameful educational environments? To achieve the objective, the following eligibility criteria were defined:

- **Inclusion criteria**: *i)* Primary/empirical studies about flow experience analyses in gameful environments; *ii)* studies published in the five years (2018-2022).
- **Exclusion criteria**: *i)* Secondary or tertiary studies (*e.g.*, other literature reviews), *ii)* gray literature (*i.e.*, non-peer-reviewed studies), *iii)* non-English written studies, *iv)*

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purely theoretical studies (*i.e.*, studies that do not analyze the flow experience) and *v)* scales development/validation (*e.g.*, studies that only propose a scale to analyze the experience of flow).

These criteria made it possible to select only primary studies that presented some type of analysis of the participants' flow experience.

In this study, following the example of recent secondary studies in this field [14]–[16] we are using the Scopus[1] database, which includes all the main databases in the area (*e.g.*, ACM Digital Library, IEEE Xplorer, Science Direct, and Springer Link), allowing general identification of published studies, while avoiding the acquisition of duplicate studies.

Next, we defined the search strategy (*i.e.*, search string). We used the method PICOC (Population, Intervention, Comparison, Outcomes, and Context) [17] to define our search string. Thus, the following PICOC was defined:

- **Population**: studies that analyze students' flow experience in gameful environments;
- **Intervention**: methods used to analyze students' flow experience in gameful environments;
- **Comparison**: not applicable, since the purpose of this study is to provide a systematic literature review;
- **Outcomes**: aspects related to students' flow experience analysis in gameful environments;
- **Context**: studies about students' flow experience analysis gameful environments.

After applying the PICOC method, the search string was defined (see Table I). The generated search string was validated by comparison with topics presented in recent studies in this field by two different experts. We limited the publication time (2018-2022). This limitation allowed access to more recent studies, which at the same time covered the current literature on the subject. We also automatically (*i.e.*, in the string itself) removed secondary/tertiary studies, gray literature, and non-English written studies.

TABLE I
SEARCH STRING

(TITLE-ABS-KEY (flow) AND TITLE-ABS-KEY (gamif*) OR TITLE-ABS-KEY (“game-based”) OR TITLE-ABS-KEY (“serious game*”) OR TITLE-ABS-KEY (exergame*) OR TITLE-ABS-KEY (“simulation game*”) OR TITLE-ABS-KEY (“persuasive techno*”) OR TITLE-ABS-KEY (games-with-a-purpose) OR TITLE-ABS-KEY(“educational game*”) OR TITLE-ABS-KEY(“game* for lear*”) OR TITLE-ABS-KEY (“learning game*”)) AND (EXCLUDE (DOCTYPE,“cr”) OR EXCLUDE (DOCTYPE,“ch”) OR EXCLUDE (DOCTYPE,“re”) OR EXCLUDE (DOCTYPE,“er”) OR EXCLUDE (DOCTYPE,“bk”) OR EXCLUDE (DOCTYPE,“ed”)) AND (LIMIT-TO (PUBYEAR,2022) OR LIMIT-TO (PUBYEAR,2021) OR LIMIT-TO (PUBYEAR,2020) OR LIMIT-TO (PUBYEAR,2019) OR LIMIT-TO (PUBYEAR,2018)) AND (LIMIT-TO (LANGUAGE,“English”))

In the *first step* an automation tool (*i.e.*, Parsifal[2]) was used to identify duplicate studies. In the *second step*, title and abstract reading were used to define whether a study met the review's inclusion criteria. In the *third step*, full article reading was used to collect data. No automation tool was used in the data collection process.

[1]https://www.scopus.com/

[2]https://parsif.al/

III. RESULTS

From a total of 668 studies identified in the initial analysis, in the end, 131 studies were identified as eligible and included in the review (full reader for data extraction). The vast majority of studies have analyzed the flow experience through scales (*i.e.*, scale (118 studies), interview (three studies), inventory (two studies), scale and data logs together (one study), scale and physiological data together (one study), open-ended question (one study), facial recognition (one study), EEG (one study), data logs (one study), not specified (three studies)). Among those who analyzed using a scale, it is possible to note that there is a pattern in the instrument used, with most of them using previously analyzed instruments (analyzed in terms of psychometric properties). Few studies have invested in using other instruments, such as behavioral data (data logs) or facial recognition. However, some studies have already been paying attention to this possibility.

IV. DISCUSSION

In the field of gameful educational environments, a significant challenge lies in identifying and understanding students' flow experience. However, the current state-of-the-art remains elusive, posing a substantial obstacle in this area of research. To address this challenge, we conducted a systematic literature review that focuses specifically on the analysis of flow experience within gameful educational environments. The systematic literature review results provide valuable insights into the current state of flow experience analysis within gameful educational environments.

Our finding reveals a predominant reliance on the use of scales for measuring students' flow experience within gameful educational environments. This result also corroborates recent finds in correlated fields [14], [18] and indicates a widespread practice among researchers to employ established measurement tools that capture various aspects of flow, such as the dimensions of challenge, skill, and immersion (also confirm results from secondary studies in similar fields). However, it is important to consider the limitations of relying solely on scales, as they may not fully capture the nuanced nature of flow experiences. Future research should explore complementary methodologies, such as qualitative approaches or mixed-method designs.

Another noteworthy result is the emphasis on using psychometrically validated instruments in flow experience analysis. This finding highlights the maturity of the community in employing rigorous measurement techniques to ensure the reliability and validity of flow assessments. By adhering to established psychometric standards, researchers enhance the credibility of their findings and contribute to the advancement of knowledge in the field. This result also highlights that the community can invest in new ways to improve flow experience analysis in gameful educational environments [1], [14], [18].

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Overall, these results shed light on the current landscape of flow experience analysis in gameful educational environments. They underscore the need for further exploration of alternative measurement approaches, the consideration of context-specific factors, and ongoing refinement of measurement standards. By addressing these research gaps, future studies can deepen our understanding of the intricacies of flow experiences in educational settings, inform the design of effective interventions, and contribute to the enhancement of gameful educational practices.

A. Threats to validity and limitations

During this study, several inherent threats to validity and limitations associated with the research approach became apparent. Primarily, there is a possibility that certain important studies were unintentionally excluded from the review due to reasons such as non-indexing or unavailability within the selected databases. To address this threat, we employed a comprehensive database that encompasses prominent publications within the field, ensuring the inclusion of studies that may have been present in other databases. Additionally, to access the full versions of identified studies that were initially unavailable, we proactively reached out to the respective authors to share the publication. It is worth noting that most of the research process relied on manual efforts by researchers, rather than the utilization of automated tools. Consequently, it is acknowledged that errors may have occurred during both phases of the project, such as potential inaccuracies during data extraction. Furthermore, it should be acknowledged that the review was conducted with a cutoff point in 2022, which implies that more recent studies (published from 2023) may not have been incorporated, thus highlighting a potential limitation in capturing the most up-to-date findings within the field.

V. Concluding Remarks

The systematic literature review presents an overview of flow experience analysis in gameful educational environments. The findings highlight the predominant use of scales to measure students' flow experience. Furthermore, the emphasis on psychometrically validated instruments underscores the importance of robust measurement tools. Examining the existing research, we offered insights into the fields of educational technologies, gameful design in education, and Positive Psychology. These findings serve as a foundation for future investigations and contribute to the advancement of our understanding of flow experience analysis within gameful educational environments. We aim as a future study to extend this review by examining new aspects related to the flow experience identification in gameful educational systems.

Appendix

The study dataset can accessed from this link: https://osf.io/27xnc/.

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